

DODÉKATHLOS

Design Brief for the Final Project

Digital Heritage and Multimedia (I.C.) A.A. 2025/2026, held by Professor

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ABSTRACT

The project that will be presented is called Dodékathlos and it is a hybrid board game designed as the Final Project of the Digital Heritage and Multimedia course. The institution thought to be the client of the game is the Archeological Museum of Olympia, one of the most important sites for ancient archeology. The chosen output of the design process is a game because it is a powerful tool for involving a large group of people, and the amount of interaction that it brings develops a strong environment-visitors bond. The Design Brief includes the documentation of the design process, supported also by the images of the personas and of the swimlane flowchart. The project aims to help the institution to enrich the educational offerings for middle school students and to include Human-Computer Interaction that is lacking at the moment.

KEYWORDS

Hybrid game, cultural heritage, human-computer interaction, visitor feedback, museum modernization, Greek mythology, archeology.

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1. INTRODUCTION

1.1 Context of the project

The Archeological Museum of Olympia is the chosen institution for setting this game, in particular the inspiration has been the metopes of the Temple of Zeus, that represent all the Twelve Labours of Hercules. Celebrated for being one of the biggest and most important Sanctuary of Ancient Greece, the Temple was home of Ancient Olympic Games and it

housed the famous statue in gold and ivory of Zeus made by Phidias, considered to be one of the seven wonders of the ancient world.

1.2 Main pain points the project aims to solve

The primary audience of the museum at the moment is not children and teenagers: the museum has indeed some solutions for school groups but the little visitors that come with their families on holiday find themselves disoriented and they lose attention really quickly. Our game is designed to entertain them and make them learn about what they are looking at, in a fun way.

Another thing is that the museum lacks interaction between humans and the digital world, so the visit is quite slow and old-style, just looking at the manufactures and at the labels on the wall. This project aims to expand the museum resources beyond that including a mobile app and the usage of a smartphone within it, bringing the works of art to life.

In addition to that, it is to say that the museum welcomes many visitors: many times this results in crowds that obstruct the view or the passage between a room and the other, and eventually an annoying noise. Thanks to this game we hope to disperse the crowd, intent on playing in a quiet spot inside a dedicated space inside the museum or outside, where a large park can accommodate the players.

2. RELATED WORKS AND BENCHMARK

2.1 Existing references, comparable projects

Existing solutions have been subdivided into three categories: games on the same topic (Hercules, Greek mythology, etc.), games with dynamics we took inspiration from, and technologies already implemented and used for other projects. For the first category we individuated *Elysium*, *Mythomakya*, *Micro Heroes: Hercules* and *Hercules and the 12 Labours*; we have been inspired by *Dungeons and Dragons* and *Cluedo* for some game dynamics.

2.2 Gap statement

Although Greek mythology is fundamental in many games, just a few of them present Hercules as their main character and almost none of them tells something about its myth, in fact Hercules is treated like a common strong hero, and nothing more. In addition to that, the archeological aspect, the material sources, is discarded: our game is specifically thought to teach about that during the match.

3. METHOD AND DESIGN PROCESS

3.1 Design method used

The design process was supported by PERCEIVE, a co-design tool. Using this tool the team chose the primary and the second audience, goals, motivations, devices and eventual barriers. Information for institutional goals, star assets, and subjects of the game were retrieved on the online site of the Archeological Museum of Olympia. A swimlane flowchart was then translated into a diagram (see [Figure 5 in Section 8.2](#)). Sketches have been produced before prototyping for having an idea of the colours, the graphics and dimensions of the game components, as well as a wireframe for having an idea of the app (see [Figure 1 below](#)). In the end, we produced a Twine story that is highly representative of the attended User Experience and a first prototype of the app, to see how it works and how it looks like.

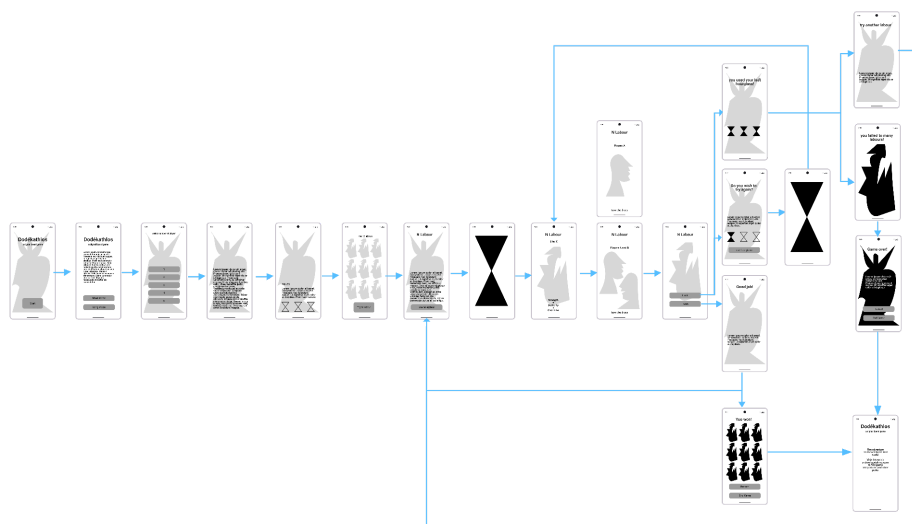


Figure 1. Wireframe of the app.

3.2 Data and materials used

The game involves the use of cards, pawns, a board, some tokens, a mobile app and special dice with six faces but on one of those there is a Lightning Bolt. In the app, actual reproductions of works of art collected in the museum will be shown. All the technical tools used in the design process will be found in the [References section](#) of this document.

4. THE INSTITUTION

4.1 Name and description

The proposed institution for this project is the Archeological Museum of Olympia located in Olympia, Greece. The location includes the archeological site that was discovered in 1766 by

the British traveler Richard Chandler and brought back to life thanks to the German Archeological Institute starting from 1875. The exhibition of the museum collects the manufacts that have been discovered in the excavations nearby in order to narrate the story of the Sanctuary of Zeus from prehistoric times to the early Christian centuries.

4.2 Star assets

The star assets of the museum are the fame that it has and the importance of the exhibition itself for the History of Art and History in general. In fact, this institution is well-known worldwide: visitors and tourists come from all over the world, attracted by the sculptures, the excavations, the old stadium, the olive tree that someone into sport recognizes as the place where the Olympic Flame is lit up still now.

4.3 Institutional goals

The institutional goals that we have detected involve more interaction with visitors and to enlarge the number of educational activities. Despite being famous and important, the visit is confined to “look and move on”: the visitors cannot do anything more than observe the manufacts and the corresponding label. Connected to this, there are some activities for school groups but their primary audience are children of elementary school, while our project is thought for slightly older middle school teenagers.

5. THE AUDIENCE

5.1 Primary audience

The primary audience chosen is students aged 11 and over that visit the exhibition with their families or with their school group. The goals are to entertain them in a place that could be boring and to promote social interaction, reinforcing relationships with their schoolmates or simply having a good moment with their relatives. However, this audience could experience some barriers like low self esteem, or they could find irrelevant what the museum or our game has to offer: this is a critical age for many reasons and we would like to involve everyone and raise interest. Nevertheless students are digital natives and they will not have problems handling devices such as computers, tablets and smartphones, and they likely have capacity in computer gaming, softwares and mobile apps.

5.2 Secondary audience

The secondary audience that we hope will choose to play our game is whoever is interested in Greek mythology, Archeology or History, so they will be motivated by personal relevance and academic interest. This project offers digital insights of the museum, in order to add

something to what enthusiasts may have already learned. Despite this, they could still find this game irrelevant because for them it is too simple or childish. Another point is that it has a cost too high and they cannot afford it: visiting the museum could be a costly expense for someone, e.g. a family or those who come from afar, so adding also this extra could be an obstacle. There are no special capabilities to learn: computer softwares, gaming mechanics and mobile apps are sufficient. As it was for the primary audience, also the secondary audience is able to handle devices like computers, tablets and smartphones. For this audience, the goals consist of feedback between visitors and environment and stimulate emotional responses: the game should add something to the plain visit and make good memories about the museum while interacting with the environment and the other players.

5.3 Personas



Figure 2. Persona no. 1

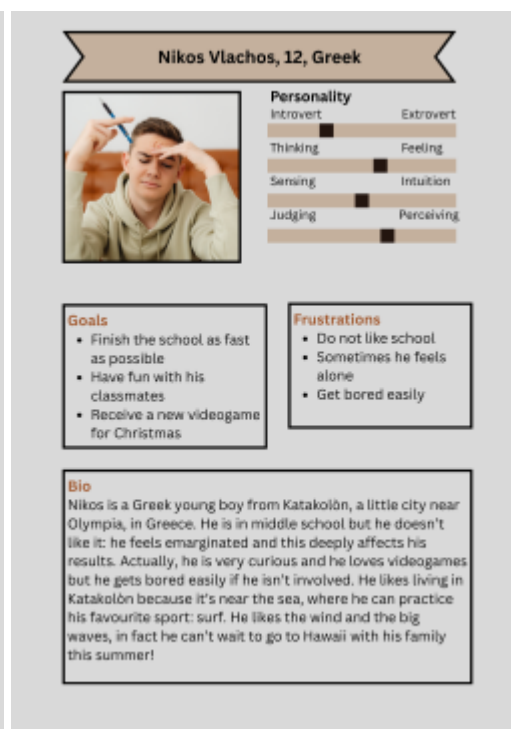


Figure 3. Persona no.2

6. THE CONCEPT

Dodékathlos is a hybrid cooperative board game. It can be played by people from 11 years old in a group of a maximum of five people and it combines the usage of traditional board games tools, like a board, dice and cards, with a mobile app to be downloaded on a smartphone. The app is mandatory in order to be guided properly throughout the game.

Players must work together to perform the mythological Labours of Hercules by choosing between five characters: Hercules (mandatory if there is only one player), Iolaus, Athena,

Theseus or Hermes. All these characters have some special abilities, called Core Attributes, that would help to achieve victory. Each Labour has a Difficulty Class and three Core Attributes that are necessary to win that Labour. Players can choose between two modalities: short (six tasks to complete) and long (twelve tasks to complete). The scope of the game is to collect at least four Labours in the short modality or at least eight Labours for the long modality. The Nike of Paionios has been chosen as the guide of the game because it is one of the jewels in the museum's crown, right in front of the visitors as they walk into the museum and at the center of the room where the metopes are collected.

7. THE MECHANICS

The game revolves around five Core Attributes: Strength, Dexterity, Intelligence, Wisdom, and Charisma (see [Figure 4](#) for the specific points assigned to each character):

Hero	Strength	Dexterity	Intelligence	Wisdom	Charisma
Heracles	5	3	1	2	4
Iolaus	3	5	2	3	2
Athena	2	3	5	4	1
Theseus	4	2	2	2	5
Hermes	1	4	4	1	5

Figure 4. The Core Attributes and their assigned points

To beat a Labour's Difficulty Class, players must add the sum of the three Core Attributes required by that specific Labour to their physical dice rolls. Life inside the game is a restricted resource managed directly by the app. Players start with three Hourglasses for each Labour but simply entering a new Labour costs one Hourglass. If a dice roll fails, players can spend one additional Hourglass to retry immediately. A Labour can be attempted a maximum of three times (costing a total of three Hourglasses) before it results in automatic defeat. Scanning physical QR codes at the museum permanently unlocks Special Perks assigned to the character cards. These act as persistent gameplay tools designed to facilitate victory in future matches.

8. USER FLOW, INTERACTION AND HYBRID SYSTEM

8.1 User flow

Players open the board and download the app. They indicate the number of players, choose if playing the short or the long mode, and select which character they want to be. Nike will then introduce the myth of Hercules. Then, explains how Hercules won the specific Labour that players are facing at that moment. For resolving a Labour, the player in charge taps one Hourglass on the app and rolls the physical dice. Pay attention to the Lightning Bolt drawn on the dice: when this comes out the player in charge picks a card from the Peril&Aid Deck. If the challenge is successful, players tap “Victory” on the app and place a Gold Coin on the board. In the opposite case, if players fail the challenge three times, they must tap “Defeat” in the app, place a Black Coin on the board and move to the next Labour. At the end of the game, in both cases of victory or defeat, Nike announces that the adventure continues in the real world, prompting players to find QR codes inside the physical museum.

8.2 Swimlane flowchart

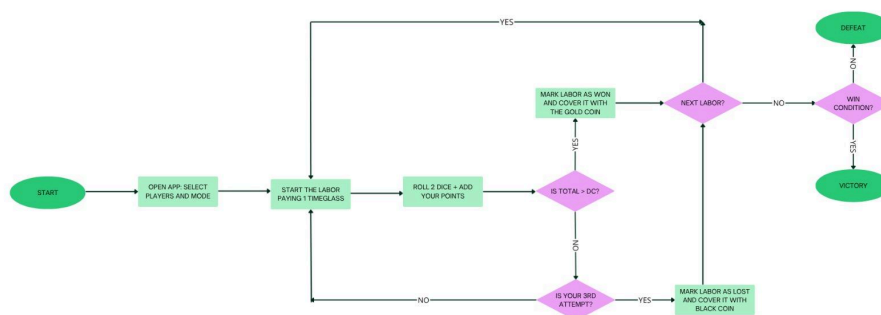


Figure 5. Diagram of the user flow with the mechanics of actors, responsibilities, decisions and failure states.

8.3 Interaction

Interaction is developed in the form of a mobile app where users can find a guide, the Nike of Paionios, that will lead the players through the game and she will explain the myth and the metopes, too. In addition to that, QR codes will be posted near the metopes as an expansion of the game in the manner described above.

9. USER JOURNEY

Discovery: the game will be promoted in the museum's shop and on the museum's social media channels for reinforcing the visitor-institution relation. The game introduces remote players to the Archaeological Museum of Olympia for the first time, sparking curiosity.

Onboarding: players realize they can choose their environment: they can play directly inside the museum, outdoors in the surrounding park, or back at home. The app is intuitive, immediately easing players into the game with a smooth introduction to the core scoring and Hourglass time mechanics.

Capture/Interaction: the Nike that comes to life is a nice surprise for the players because it is rare seeing a statue that speaks and moves. Users will find no difficulties in understanding the game mechanics, especially for the functionalities of attempts (the Hourglasses) and scoring.

After experience: win or lose, players finish the session feeling satisfied but craving more. Nike announces that the adventure continues in the real world. This creates a powerful incentive for players to visit (or return to) the Archaeological Museum of Olympia to scan QR codes and permanently unlock character perks for their next playthrough at home.

10. PAIN POINTS AND INNOVATION

The first pain point that has been raised in Section 1 is the fact that the museum does not involve directly the adolescent and youth demographic. Our project is specifically designed for them, in fact we chose teenagers and middle school students as the primary audience. We expect that they will feel at ease and engaged while learning new things and developing a passion for History and Archaeology. By adding interactive elements inside the museum and bringing works of art to life inside a smartphone, we resolve the problem of a lack of interaction that we have encountered. Visitors and players would be surprised and enchanted by the technology. Thanks to the fact that this game could be played in a separated space of the museum, e.g. the courtyard, it could be an helpful tool to disperse the crowd that inevitably forms because the museum welcomes hundreds of tourists every day. In this way, visitors can proceed at a more leisurely pace and enjoy the artworks with greater tranquility.

11. CONCLUSION

Dodékathlos is a project that aims to help the Archeological Museum of Olympia in those aspects where something is missing. The game will contribute in modernizing the museum and improving the visit toward an involving relationship between the environment and the

visitors. Apart from that, both an educational and a social aspect are touched: players will learn something more specific about archaeological discoveries and Greek myths. Anyone will be involved in the game with a specific role, so nobody will be excluded and everyone will bring a special contribution in order to win. Limitations of the project relies on the fact it will be promoted only by the museum's channels, so maybe it would not reach a big audience worldwide. In addition to that, those already well-versed in the subject matter may find the game lacking in depth and, consequently, boring. The team will direct future efforts in expanding the game with other works of art inside the museum and adding some technologies inside the building (like screens or on-site videogames), apart from promoting at least this early version to the widest possible audience.

ACKNOWLEDGMENTS AND TEAM ROLES

Camilla Neri - Prototype designer and technical lead: prototyping, testing, technical feasibility, software and hardware integration. Visual designer: visual design, art direction, user interface layout, aesthetic cohesiveness.

Angela Paganini - Narrative: storytelling and theme, script and dialogues. Twine story. Writer of the Design Brief.

Susanna Valentini - Research: content strategy, information architecture, multimedia data production. Experience and Interaction designer: User flow, User journey, interaction design.

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WEB LINKS

- Archeological Museum of Olympia: <https://ancientolympiamuseum.com>
- Canva: <https://www.canva.com>
- Figma: <https://www.figma.com/>
- Adobe Photoshop: <https://www.adobe.com/>
- Procreate: <https://procreate.com>
- Bootstrap: <https://getbootstrap.com>
- Image of the Hermes of Praxiteles:
http://www.parosweb.com/guide/about_paros/white-marble/greek-statues/hermes-praxiteles.html
- Image of the Nike of Paionios (on the website):
https://upload.wikimedia.org/wikipedia/commons/thumb/6/61/Nike_of_Paionios%2C_Olympia_Archaeological_Museum_%2816309967616%29.jpg/500px-Nike_of_Paionios%2C_Olympia_Archaeological_Museum_%2816309967616%29.jpg
- Image of the metope of the Nemean Lion:
<https://quizlet.com/gb/790651258/nemean-lion-metopes-of-the-temple-of-zeus-at-olympia-flash-cards/>
- Image of the Nike:
https://ancientolympiamuseum.com/wp-content/uploads/2025/10/M_127.png